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Problem

ESCAPE FIXED LABEL SETS

- Supervised vision learns predetermined categories and needs new labeled examples for every additional concept.
- Internet image-text pairs offer broad supervision, but the learning task must scale to noisy natural language.

2

Data & encoders

400M WEB PAIRS

- A dataset of 400 million image-text pairs trains an image encoder (ResNet or ViT) and a Transformer text encoder from scratch.
- Each encoder maps its modality to one shared normalized embedding space; inference can cache text prototypes.

3

Objective

MATCH THE RIGHT CAPTION

- Within each batch, all N by N image-text cosine similarities are scored and a learned temperature scales logits.
- Symmetric cross-entropy predicts the correct caption for every image and the correct image for every caption.

CLIP

Learning Transferable Visual Models

Radford et al. | arXiv:2103.00020v1

Pretrain image and text encoders to identify matching web pairs, then turn natural-language class descriptions into zero-shot visual classifiers.

contrastive learning

image-text

zero-shot

4

Zero-shot transfer

CLASSES BECOME PROMPTS

- Class names are embedded inside natural-language templates such as 'a photo of a {label}'.
- Prompt ensembling and dataset-aware wording improve robustness without updating model weights.

5

Evidence

MORE THAN 30 DATASETS

- CLIP transfers to OCR, actions, geolocation and fine-grained recognition; zero-shot ImageNet matches the original supervised ResNet-50.
- The largest model is often competitive with task-trained baselines and is more robust to several natural distribution shifts.

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Meaning & limits

- CLIP made open-vocabulary recognition practical and became a common semantic backbone for multimodal generation and retrieval.
- Web supervision imports social bias; counting, fine-grained distinctions and abstract tasks remain weak, and prompts affect results.